

MAT 1043 04A

Quantitative Reasoning

Course Syllabus - Fall 2026



Course Information

Course Description: Prerequisite: Satisfactory performance on a placement examination; this course is designed primarily for the liberal arts major to satisfy the Core Curriculum Mathematics requirement. Topics may include logic; proofs; deductive and inductive reasoning; number theory; fundamentals of statistics; basic statistical graphs; causal connections; financial management; functions; linear graphs and modeling; exponential growth and decay; logarithms; fundamentals of probability; fundamentals of geometry; and basic ideas from trigonometry, calculus, and discrete mathematics. (Formerly MTC 1043. Credit cannot be earned for both MAT 1043 and MTC 1043.) May apply toward the Core Curriculum requirement in Mathematics. Generally offered: Fall, Spring, Summer. Generally Scheduled Location: Downtown Campus, Main Campus, Online/Internet. Course Fee: LRC1 \$12; LRS1 \$46.20; STSI \$21.60; DL01 \$75.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 08/19/2026-12/11/2026

Campus: Main Campus

Location: [MH 3.04.20](#)

Time(s): MWF 12:00 PM - 12:50 PM

Learning Goals

At the end of the course, you will be able to:

- Communication goal:
Outcome
You will interpret and communicate quantitative information and mathematical and statistical concepts using language appropriate to the context and intended audience.
- Problem Solving goal:
Outcome
You will make sense of problems, develop strategies to find solutions, and persevere in solving them.
- Reasoning goal:

Outcome

You will reason, model, and make decisions with mathematical, statistical, and quantitative information.

- **Evaluation goal:**

Outcome

You will critique and evaluate quantitative arguments that utilize mathematical, statistical, and quantitative information.

- **Technology goal:**

Outcome

You will use appropriate technology in a given context.

Communicate with Me



Instructor Name: Garrett DeLong

Department
Mathematics

Office Location
FLN 1.02.04

Student Hours
MWF 1:00 PM to 2:00 PM (after class) or by appointment

Email Address: garrett.delong@utsa.edu

Preferred Method of Communication
Email

Instructor Name: Stephen Lee

Department
Mathematics

Student Hours
N/A

Email Address: stephen.lee@utsa.edu

Preferred Method of Communication
Email

About Me & My Teaching Philosophy

I am a graduate teaching assistant for the department of math. This is my second year in the graduate program and seventh year as a student at UT San Antonio. Some of my hobbies are acoustic guitar, crochet, TTRPGs, photography, sewing, and video games. My favorite games (in no order) are Cyberpunk 2077, Dishonored, Dread Delusion, Fallout: New Vegas, ULTRAKILL, Skyrim, and Terraria. My favorite bands (in order) are TOOL, Pink Floyd, My Chemical Romance, Paramore, and Joji. I am excited to learn about y'all and be a part of your math education.

My teaching philosophy is defined by empathy and equity. I believe that every student is capable of doing math and succeeding academically given support and with internal motivation. I am also an advocate of the Rights of the Learner (Crystal A. Kalinec-Craig) which are (1) You Have the Right to Be Confused, (2) You Have the Right to Claim a Mistake, (3) Have the Right to Speak, Listen and be Heard, and (4) You Have the Right to Write, Do, and Represent Only What Makes Sense to You.

Course Materials

Microsoft Office

Required/Recommended/Optional: Required

Additional Information

downloaded for free through UTSA at my.utsa.edu

Course materials will be available on Canvas.

Scientific calculator

Required/Recommended/Optional: Required

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Assessments and Assignments

Homework: Homework will be given through problem sets that are divided by Module or in a few cases by topics (that extend across multiple modules). Each problem set is intended to be the most difficult type of assignment you should do for each topic (i.e. you should probably find the problem set questions harder than test questions). You are welcome to work with others on your homework, BUT you should properly credit others if you work together with someone type of acknowledgement. Further, everyone is expected to turn in their own assignment and it should not just be a copy of someone else's or the same file. Beyond that, if you ever need more help on an assignment, please reach out to me and let me know as I am always here to help you!

Student Portfolio: We will work through the In-Class Notes during class sessions in each unit in the course and these will be provided on Canvas. You can either electronically write on a PDF version of the In-Class notes or print out the pages and submit a scan of your completed handwritten notes using the PDF pages provided to submit for your Portfolio. The Portfolio will be graded to help monitor student learning and consists of a chronological order of work completed during the course from the In-Class Notes as well as select printed pages from Preview and/or Practice assignments. Any work done on notebook paper or in Excel (.xls files) must be turned in as well. Your portfolio will be turned in online with one submission for each of the 12 Lessons. Portfolios will be graded on completeness as well as accuracy.

Mini-Projects: May include well-written essays and extensive work in Excel, are interspersed throughout this course within the class notes. The mini-projects will be assigned and submitted through Canvas as we encounter them within the existing course materials.

Exams: Three exams will be given this semester in person. Each student is required to take all three exams, in addition to the final exam. Make-up Exams may be allowed if the student contacts the instructor promptly. Make-up Exams must be scheduled within one week of the original exam date. The exam grades are averaged together and count as 25% of the overall grade.

Final Exam: The final exam will be a comprehensive final and is mandatory for all students; it counts as 20% of the overall grade.

1. The final exam in this course will be comprehensive, including all material covered during this course.
2. Students should study their previous exam reviews to prepare them for the final exam.
3. The final exam will be given in person at the end of the semester.

Expectations/Participation/Attendance:

Introduction to Math (Quantitative Reasoning) is a course that builds on previous knowledge, therefore staying up to date with the lessons according the course schedule and engaging in the activities through Preview/Practice assignments, In-Class notes, Class discussion, and Mini-projects will help you be successful in the course.

You will learn Quantitative Reasoning by actively participating in the lessons. Students enrolled are expected to participate as a learning community, by attending and being prepared for class, engaging in group activities, joining class discussions, communicating their understanding of mathematics, and explaining their work to others.

Attendance will be taken each class. You may accumulate 2 absences before your attendance grade is affected. **Additionally, if a student accumulates 2 consecutive weeks (6 classes for MWF or 4 classes for TR), or 3 cumulative weeks (9 classes for MWF or 6 classes for TR) of non-University-excused absences without communication with the instructor, then the instructor reserves the right to drop that student from the course.**

Make-Up Policy:

A course schedule is provided to keep you on page with the course. Late work will be accepted according to instructor rules and students should follow individual instructor policy regarding late work.

Exam make ups can be given but must be done so in a timely fashion and with appropriate communication and approval from the instructor. Failure to adhere to this may result in make ups not being given even if for valid reasons (i.e. you are sick on an exam day but then do not schedule a time to make up the exam).

Activities and Grading

Grades will be divided into five categories (explained below) and weighted according to the following percentages:

Grading Scheme:	Weight as a %
Problem Sets	15%
Student Portfolio	25%
Mini-project average	10%
Attendance/Participation	5%
Exams (3 total)	25%
Final Exam (comprehensive)	20%
Total	100%

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	96.5% - 100%
A	92.5% - 96.49%
A-	89.5% - 92.49%
B+	86.5% - 89.49%
B	82.5% - 86.49%
B-	79.5% - 82.49%
C+	76.5% - 79.49%
C	72.5% - 76.49%
C-	69.5% - 72.49%
D+	66.5% - 69.49%
D	62.5% - 66.49%
D-	59.5% - 62.49%
F	less than 59.5%

Course Guidelines for the Use of Generative Artificial Intelligence

To best succeed in this course, you should complete all assignments (Portfolios, Problem Sets, Miniprojects, Exams, etc) without the use of any generative AI tools (i.e. ChatGPT, Claude, Gemini, etc). As well please do not use AI to generate any content that you would try to pass off as your own in general. The purpose of these assignments is to see ***your*** thinking and understanding. We design the course policies to focus on offering you the opportunity to revise your work (and alleviate the grade stress). If you are caught or believed to be using AI generated content in graded assignments, you may be in violation of Academic Integrity principles (and brought before Academic Integrity) and may suffer punitive measures because of it.

Course Expectations & Policies

Instructor-Initiated Drops

This course uses instructor-initiated drops for students who exceed the absence and/or missed assignment limit. Therefore, up to the last day for students to withdraw from an individual course, **October 23rd**, you will be dropped for exceeding **if a student accumulates 2 consecutive weeks (6**

classes for MWF or 4 classes for TR), or 3 cumulative weeks (9 classes for MWF or 6 classes for TR) of non-University-excused absences without communication with the instructor. Students will receive at least one courtesy warning when approaching the absence/missed assignment limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. *This drop does not affect enrollment in other courses.* **Please consult the [Dropping Courses webpage](#) for further details on the process and appeals.**

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Academic Freedom

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
2. 1940 Statement of Principles on Academic Freedom and Tenure
3. Board of Regents Rule 31004; HOP Policy 4.02
4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
5. The Chicago Statement

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Due Date	Activities/Assignments	Points
	Attendance Aug to Sep	12
	Attendance Grade	100
	Exam 1	110
	Exam 1 Actual	100
	Exam 2	110
	Exam 2 Actual	100
	Exam 3	110
	Exam 3 Actual	100
	Final Exam Overall	100
	Lesson 10 Portfolio Submission	15
	Lesson 11 Portfolio Submission	15
	Lesson 12 Portfolio Submission	15
	Lesson 9 Portfolio Submission	15
	Miniproject 4	20
	Module 1 Problem Set Actual	38

Due Date	Activities/Assignments	Points
	Module 10/11 Problem Set	60
	Module 2 Problem Set Actual	47
	Module 3 Problem Set Actual	34
	Module 4 Problem Set Actual	45
	Module 9 Problem Set	34
9/6/26	Lesson 1 Portfolio Submission	15
9/6/26	Lesson 2 Portfolio Submission	15
9/6/26	Module 1 Problem Set	44
9/6/26	Module 2 Problem Set	49
9/13/26	Lesson 3 Portfolio Submission	15
9/13/26	Module 3 Problem Set	36
9/20/26	Miniproject 1	20
9/20/26	Lesson 4 Portfolio Submission	15
9/20/26	Module 4 Problem Set	48
10/4/26	Lesson 5 Portfolio Submission	15
10/4/26	Module 5 Problem Set	66
10/9/26	Lesson 6 Portfolio Submission	15

Due Date	Activities/Assignments	Points
10/9/26	Module 6 Problem Set	44
10/11/26	Miniproject 2	20
10/18/26	Lesson 7 Portfolio Submission	15
10/18/26	Miniproject 3	20
10/18/26	Module 7 Problem Set	42
10/25/26	Module 8 Part 1 Problem Set	37
11/1/26	Lesson 8 Portfolio Submission	15
11/1/26	Module 8 Part 2 Problem Set	60

The course topics will be covered according to the following plan (dates are subject to change). Additionally, there will be a more detailed schedule on Canvas.

Lesson	Section	Date	Topics
1	1.A	Week 1	Data
2	1.B	Week 1	Learning Community
3	1.C	Week 1	First-degree equation involving percentages
5	1.D	Week 2	Ranking methods
6	2.A	Week 2	Graphical Display
7	2.C	Week 2	Analyzing graphical displays
8	3.A	Week 3	Sampling
9	3.B	Week 3	Mean and Central Limit Theorem
10	3.C	Week 3	Standard Deviation
11	4.A	Week 4	Probability
12	4.B	Week 4	Conditional Probability
13	5.A	Week 5	Conversions
14	5.B	Week 5	Index numbers
15	5.C	Week 5	Weighted averages
16	5.D	Week 6	Expected Value
17	6.A	Week 6	Weighted moving average graphs
18	6.B	Week 6	Weighted moving average graphs continued
19	7.A	Week 6	Ratios and percentages
20	7.B	Week 7	Part-to-part ratios and Part-to-whole ratios
21	7.C	Week 7	Absolute change (additive reasoning) and Relative change (multiplicative reasoning)
22	7.D	Week 7	Adjusting claims and hypothesis
23	7.E	Week 8	Debt-to-income (DTI) ratios
24	7.F	Week 8	Proportional reasoning
25	8.A	Week 8	Mathematical (Linear) relationships
26	8.B	Week 9	Proportionality vs. Linearity
27	8.C	Week 9	Simple and Compound Interest (Linear and Exponential Models)
28	8.D	Week 10	Regression
29	8.E	Week 10	Piecewise Linear Function
30	9.A	Week 10	Depreciation
31	9.C	Week 11	Univariate, Bivariate, Correlation and Causation
32	9.D	Week 11	Univariate, Bivariate, Correlation and Causation
33	10.A	Week 11	Time series model of exponential growth
34	10.B	Week 12	Linear and Exponential Models
35	11.A	Week 12	Continuous Growth
36	11.B	Week 12	Carrying Capacity and Logistic Growth Rate
37	11.C	Week 13	Logistic Growth Model
38	11.D	Week 13	Complex Population Growth and Decay Models
39	11.E	Week 14	Analyzing Complex Population Growth and Decay Models
40	12.A	Week 14	Periodic Function
41	12.B	Week 15	The Sine Function

Department Policies & Resources

Student Advocate

The student advocate is a person who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor; it is highly encouraged that you first reach out to the instructor with any issues to see if they can be resolved.

To reach the department's student advocate, please email mathematics@utsa.edu and include "Student Advocate" in the subject line of your email.

Tutoring Services

UT San Antonio provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213, MAT 1223, MAT 2213, MAT 2253, MAT 1313

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule for drop-in hours and course-specific targeted sessions will be provided by instructors through Canvas.
- **Location:** [Math Gym, Main Building 0.414](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, accommodations for pregnant students, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.

- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.

Technology Requirements

- Basic computer skills.
- Computer with speakers or headphones.
- Access to Microsoft Office (downloaded for free through my.utsa.edu)
- Internet access. Wired or reliable connection recommended for online tests/assignments.

Tech Support:

- Visit the UTSA Online Learning websites for technical support: – [Tech Specs](#)
- Visit [UTSA Student Connect Computer Lab Information](#).

If you have a problem with Canvas and you have already cleared the cache and tried with a different browser, please submit a ticket to UTSA Tech Café. Be ready with your course ID and section. A screenshot of the issue will accelerate the troubleshooting process.

Contact Information for Tech Café:

- Email UTSA Tech Café at techcafe@utsa.edu
- Call UTSA Tech Café at 210.458.5555
- Visit the [UTSA Tech Café Website](#)

A screenshot of the issue you are encountering will accelerate the troubleshooting process. The UTSA Tech Café is the technology support services for UTSA students, faculty, staff and affiliate members.

Always notify your instructor about the issue you are encountering.

Ombudsperson

For any questions or issues concerning faculty or unfair/prejudicial practices affecting you, please contact Ilse Rosales Cooper at Ilse.rosalesmartinez@utsa.edu